

LACONIA, NH, USA

WMO: 726155

Lat: 43.567N

Lon: 71.433W

Elev: 166

StdP: 99.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-17.0	-27.6	0.3	-17.4	-24.0	0.4	-15.0	10.3	-2.7	9.2	-4.0	2.9	260	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.2	21.9	29.1	20.8	27.7	19.9	23.2	28.6	22.3	27.3	21.3	26.0	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.4	25.9	20.9	15.9	25.2	19.5	14.5	24.1	69.7	28.7	66.0	27.3	62.6	26.1	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.8	7.0	-22.8	33.8	3.1	1.5	-25.0	34.9	-26.9	35.8	-28.6	36.6	-30.9	37.7		
(5)			-22.7	24.7	3.1	1.0	-24.9	25.4	-26.8	25.9	-28.5	26.5	-30.8	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-6.0	-4.7	0.2	6.8	13.0	18.4	21.4	20.3	16.0	9.6	3.5	-2.1		
	DBStd	10.47	5.97	5.48	5.18	4.49	4.16	3.78	3.01	2.94	3.89	4.20	4.68	4.97		
	HDD10.0	1992	498	413	307	118	16	1	0	0	2	60	202	376		
	HDD18.3	3994	756	646	562	348	174	46	7	13	90	272	446	634		
	CDD10.0	1293	0	0	3	21	108	252	353	319	183	47	6	0		
	CDD18.3	254	0	0	0	1	8	47	102	74	21	1	0	0		
	CDH23.3	2140	0	0	4	23	125	422	844	566	147	9	0	0		
	CDH26.7	592	0	0	1	6	34	124	256	142	30	0	0	0		
(14) Wind	WSAvg	2.8	3.0	3.0	3.1	3.1	2.8	2.6	2.5	2.2	2.3	2.8	2.9	3.0		
(15) Precipitation	PrecAvg	1169	80	80	98	93	92	103	103	99	93	115	108	105		
	PrecMax	1543	163	222	178	198	245	298	201	216	242	369	157	202		
	PrecMin	777	16	38	39	11	10	31	13	22	28	16	49	44		
	PrecStd	201	33	37	36	48	52	63	45	56	50	84	26	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	12.3	20.9	27.0	30.2	32.7	32.9	32.5	29.9	24.8	18.8	13.6		
		MCWB	11.1	7.5	13.5	15.6	18.7	22.7	23.3	22.3	21.5	17.4	15.6	10.1		
	2%	DB	7.1	7.7	13.7	22.2	27.2	30.1	31.3	29.9	27.2	22.0	16.2	9.1		
		MCWB	4.3	3.8	7.8	13.1	16.7	21.0	22.3	21.5	19.7	16.2	12.3	6.7		
	5%	DB	3.8	5.1	10.0	18.0	23.8	27.6	29.1	27.9	24.8	18.9	12.9	7.1		
		MCWB	1.4	2.2	5.0	10.8	15.4	19.4	21.1	20.4	18.5	14.6	9.5	4.5		
	10%	DB	2.1	2.7	7.4	15.1	21.3	26.0	27.7	27.0	22.7	16.8	11.0	4.2		
		MCWB	0.3	0.3	3.5	8.4	14.0	18.1	20.3	19.7	17.5	12.6	7.7	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	8.6	13.5	17.5	20.5	23.9	24.7	24.0	22.8	19.4	16.4	11.3		
		MCDB	12.1	10.4	19.4	22.9	27.9	30.0	30.9	29.2	27.0	22.1	18.4	12.9		
	2%	WB	4.6	4.7	8.8	14.4	18.6	22.5	23.4	22.7	21.1	17.2	13.2	7.5		
		MCDB	6.5	7.2	12.9	20.6	24.3	28.4	29.0	27.4	25.1	20.1	15.5	8.9		
	5%	WB	1.8	2.5	5.7	11.6	16.8	20.9	22.5	21.7	19.9	15.1	10.3	4.7		
		MCDB	3.4	4.7	9.3	17.0	22.3	25.6	27.7	26.2	23.3	18.4	12.9	6.6		
	10%	WB	0.2	0.6	3.7	9.1	15.1	19.6	21.5	20.8	18.7	13.1	7.9	2.7		
		MCDB	1.7	2.6	6.9	14.5	19.9	23.9	26.3	24.8	22.0	15.9	9.9	4.3		
(35) Mean Daily Temperature Range	MDBR		8.8	9.6	9.7	10.9	11.2	10.6	10.6	10.8	10.6	9.7	8.2	7.6		
	5% DB	MCDBR	10.3	11.6	14.0	17.0	15.8	13.4	12.3	12.4	12.4	13.2	11.9	10.1		
		MCWBR	9.1	8.8	9.3	9.6	7.8	6.0	5.2	5.6	6.4	8.1	9.0	8.4		
	5% WB	MCDBR	10.0	11.1	12.8	15.5	13.1	11.7	10.7	10.5	10.3	11.5	10.6	9.8		
		MCWBR	9.2	8.9	9.4	10.0	7.4	6.1	5.2	5.4	6.2	8.0	9.1	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.289	0.316	0.348	0.379	0.427	0.445	0.414	0.366	0.331	0.308	0.283		
	taud		2.421	2.366	2.417	2.423	2.374	2.254	2.213	2.289	2.406	2.512	2.531	2.488		
	Ebn at Noon		866	911	926	918	892	847	826	841	860	853	823	823		
	Edh at Noon		74	93	101	110	119	134	139	124	102	81	68	64		
(44) All-Sky Solar Radiation	RadAvg		1.66	2.52	3.63	4.52	5.02	5.42	5.66	5.14	4.18	2.63	1.76	1.30		
	RadStd		0.12	0.20	0.23	0.51	0.60	0.50	0.39	0.28	0.26	0.26	0.18	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	N/A

Nomenclature: See separate page